SVM Notes - Hinglish - Atlas Skilltech

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SVM ek powerful algorithm hai jo best possible boundary dhundhta hai classes ke beech.

1. SVM — Introduction (Parichay)

SVM ka full form hai **Support Vector Machine**. Yeh ek supervised machine learning algorithm hai.

Simple Analogy (Seedhi Baat):

> Table par Blue aur Red marbles mixed hain. SVM ek aisi ruler (line) dhundhta hai jo inhe **maximum gap** ke saath alag kare. Yeh line itni best jagah ho ki naye marble bhi sahi side par padein.

Real World Examples:

Application	Input	Output
Email Filter	Email ka text	Spam / Not Spam
Medical Diagnosis	Tumor measurements	Cancer / Safe
Face Recognition	Pixel values	Person A / Person B
Credit Card Fraud	Transaction data	Fraud / Legitimate
News Classification	Article text	Sports / Politics / Tech

SVM vs KNN vs Logistic Regression:

Property	SVM	KNN	Logistic Regression
Training	Slow	Nothing	Fast
Prediction	Fast	Slow	Fast
Non-linear data	Kernel trick	Distance-based	Polynomial features
High dimensions	Excellent	Poor	Good

Property	SVM	KNN	Logistic Regression
Interpretability	Low	Medium	High
Memory	Small (SVs only)	Large (all data)	Small

SVM kab use karein:

- Dataset medium sized ho (not millions of rows)
- Features zyada hon (image, text data)
- Clear margin separation chahiye
- Non-linear boundaries ki zaroorat ho

2. Hyperplane aur Margins

Hyperplane kya hota hai?

- **2D mein:** Ek line `ax + by + c = 0`
- **3D mein:** Ek plane
- **nD mein:** (n-1) dimensional surface

Hyperplane dono classes ko divide karta hai. SVM woh hyperplane dhundhta hai jo **maximum margin** deta hai.

Margin kya hota hai?

Margin = Hyperplane ke dono taraf woh buffer zone.

```
\text{Margin Width} = \frac{2}{||w||}
```

Jahan `w` = weight vector (normal to hyperplane). **Margin maximize karna = ||w|| minimize karna.**

SVM Optimization Problem:

```
\min_{w, b} \frac{1}{2} ||w||^2
```

Subject to:

```
y_i(w \cdot x_i + b) \geq 1 \quad \forall i
```

Support Vectors

> Woh data points jo margin boundary par hote hain. Sirf yahi points SVM model ko define karte hain. Baaki sab points remove karo — model same rahega!

Hard Margin vs Soft Margin

Type	C Value	Behavior	Problem
Hard Margin	C bahut bada	Zero tolerance for mistakes	Noisy data par fail
Soft Margin	C chota	Kuch mistakes allow	Real data ke liye better

C Parameter (Regularization)

```
\min_{w, b, \xi} \frac{1}{2}||w||^2 + C \sum_{i=1}^{n} \xi_i
```

- **C bada (e.g. 1000):** Tight margin, low tolerance → overfitting risk
- **C chota (e.g. 0.01):** Wide margin, high tolerance → underfitting risk
- **C = 1:** Balanced (good starting point)

Analogy: C = Kitna strict teacher hai exam mein. Bahut strict = rote learning (overfit). Bahut lenient = kuch nahi seekha (underfit).

3. Kernel Intuition

Problem: Non-Linear Data

Kya hoga jab ek straight line se data separate nahi ho sakta?

Example: Circle ke andar ek class (Benign), bahar doosri class (Malignant). Koi bhi straight line inhe separate nahi kar sakti!

Kernel Trick ka Magic

Analogy — Marble Example:

> Maan lo floor par Blue aur Red marbles mixed hain — Blue wale circle ke andar, Red wale bahar. Seedha line? Impossible!
>
> Ab ek paper sheet se table thoda hila do (3D mein lift karo): Blue wale upar uthenge, Red wale neeche rahenge. Ab ek flat surface (plane) se alag kar sakte hain!
>
> Yahi Kernel Trick karta hai — bina actually higher dimension mein jaaye, wahan ka calculation karta hai.

4 Kernels — Kab Kya Use Karein

Kernel	Real World Use	Data Type
Linear	Text classification, Spam detection	High-dim, linearly separable
RBF (Gaussian)	Medical, Image, General	Non-linear, circular boundaries
Polynomial	Image processing, Computer Vision	Polynomial shaped boundaries
Sigmoid	Neural network alternative	Binary, NLP tasks

Real World Example — RBF Kernel:

> **Face Recognition:** Ek person ki alag alag angles ki photos classify karna. Linear line? Impossible (thousands of pixel features). RBF kernel? Complex boundary banata hai pixels ke space mein. Facebook/Google Photos mein yahi hota hai!

Gamma (γ) Parameter — RBF ka Remote Control

- **γ bada (e.g. 10):** Har point ka sirf uske pass wale par influence → Very jagged boundary → Overfit
- **γ chota (e.g. 0.001):** Har point ka door-door tak influence → Very smooth boundary → Underfit
- **γ = 'scale' (default):** Automatically choose karta hai → Best starting point

4. Kernel Mathematics

Linear Kernel

```
K(x_i, x_j) = x_i \cdot x_j = \sum_{k=1}^{n} x_{ik} \cdot x_{jk}
```

Seedha dot product. Koi transformation nahi. Fast aur simple.

RBF Kernel (Radial Basis Function / Gaussian)

```
K(x_i, x_j) = \exp\left(-\gamma \|x_i - x_j\|^2\right)
```

Jahan `gamma > 0`. Do points paas hain to K high, door hain to K ~0.

Yeh kya karta hai: Infinite-dimensional space mein transformation, bina wahan actually gaye!

Polynomial Kernel

```
K(x_i, x_j) = (\gamma \cdot x_i \cdot x_j + r)^d
```

Jahan: d = degree, r = coef0 parameter.

- d=2: Quadratic boundary
- d=3: Cubic boundary

Sigmoid Kernel

```
K(x_i, x_j) = \tanh(\gamma \cdot x_i \cdot x_j + r)
```

Neural network ke tanh activation jaisa.

Decision Function

SVM ka final prediction:

```
f(x) = \text{sign}\left(\sum_{i \in SV} \alpha_i y_i K(x_i, x) + b\right)
```

Jahan:

- `SV` = Support Vectors
- `alpha_i` = Lagrange multipliers (learned during training)

- `y_i` = class labels (+1 or -1)
- `b` = bias term

Parameter Comparison

Parameter	Kernel	Effect	Range
C	All	Regularization	0.001 to 1000
gamma	RBF, Poly, Sigmoid	Boundary complexity	'scale', 0.001 to 10
degree	Polynomial	Polynomial order	2, 3, 4, 5
coef0	Polynomial, Sigmoid	Bias in kernel	0 to 1

5. SVM Classification — Implementation

Complete Pipeline

```
from sklearn.svm import SVC
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import classification_report, ConfusionMatrixDisplay
import matplotlib.pyplot as plt

# 1. Data Load
import pandas as pd
df = pd.read_csv('data.csv')
X = df.drop('target', axis=1)
y = df['target']

# 2. Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

# 3. ZARURI: Feature Scaling!
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
# 4. SVM Model
svm = SVC(
    kernel='rbf',       # Linear, rbf, poly, sigmoid
    C=1.0,              # Regularization
    gamma='scale',      # RBF/Poly bandwidth
    probability=True,   # ROC curve ke liye
    random_state=42
)
svm.fit(X_train, y_train)

# 5. Evaluate
y_pred = svm.predict(X_test)
print(classification_report(y_test, y_pred))

# 6. Confusion Matrix (Naya Tarika)
ConfusionMatrixDisplay.from_estimator(svm, X_test, y_test)
plt.show()
```

GridSearchCV — Best Parameters

```
from sklearn.model_selection import GridSearchCV

param_grid = {
    'C': [0.1, 1, 10, 100],
    'gamma': ['scale', 'auto', 0.001, 0.01, 0.1],
    'kernel': ['rbf', 'linear']
}

grid = GridSearchCV(SVC(), param_grid, cv=5, scoring='accuracy', n_jobs=-1)
grid.fit(X_train, y_train)

print('Best Params:', grid.best_params_)
print('Best Score:', grid.best_score_)
```

Real World Context

> **Spam Detection:** Input: Email ka TF-IDF vector (10,000+ features) SVM linear kernel yahan best kaam karta hai kyunki:
>
> - High dimensional data
> - Linear separation (spam words clearly different hote hain)
> - Fast prediction (real-time email filtering) Gmail, Outlook sab SVM variants use karte hain!

6. SVM Regression (SVR)

SVR kya hai?

SVR = Support Vector Regression. SVM ka regression version.

Classification mein: Hyperplane dono classes ko separate karta hai. **Regression mein:** Hyperplane predictions ke liye ek **epsilon tube** banata hai.

Epsilon Tube (ε-tube)

```
|y_i - f(x_i)| \leq \varepsilon
```

> Jo points tube ke andar hain — no penalty (thoda error allowed). Jo points tube ke bahar hain — penalize karo (slack variable ξ).
>
> **Analogy:** Highway ka ek lane — jab tak car lane mein hai, koi baat nahi. Lane se bahar nikli to penalty!

SVR Objective Function

```
\min \frac{1}{2}||w||^2 + C \sum_{i=1}^{n}(\xi_i + \xi_i^*)
```

Subject to:

```
y_i - (w \cdot x_i + b) \leq \varepsilon + \xi_i
```

```
(w \cdot x_i + b) - y_i \leq \varepsilon + \xi_i^*
```

SVR sklearn Code

```
from sklearn.svm import SVR
from sklearn.metrics import mean_squared_error, r2_score
import numpy as np

svr = SVR(
    kernel='rbf',
    C=100,
    epsilon=0.1,  # Tube ki width
    gamma='scale'
)
svr.fit(X_train_scaled, y_train)

y_pred = svr.predict(X_test_scaled)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
r2 = r2_score(y_test, y_pred)
print(f'RMSE: {rmse:.2f}')
print(f'R2 Score: {r2:.4f}')
```

SVR Real World Uses

Application	Input Features	Predict
House Price	Location, Size, Rooms	Price (Rs.)
Stock Forecast	Historical prices, Volume	Next day price
Energy Demand	Temperature, Time, Season	kWh demand
Traffic Prediction	Historical flow, Time	Vehicle count

7. Real World Applications + Cheat Sheet

Industry mein SVM ka Use

Industry	Application	Kernel Used	Why SVM
Healthcare	Cancer detection, Gene classification	RBF	High accuracy, medical imaging
Finance	Credit scoring, Fraud detection	Linear, RBF	Reliable, handles imbalance
NLP/Text	Spam filter, Sentiment analysis	Linear	High-dimensional text data
Computer Vision	Face recognition, Object detection	RBF, Poly	Non-linear pixel boundaries
Bioinformatics	Protein classification	RBF	Small samples, many features
Security	Intrusion detection	RBF	Anomaly boundaries

Quick Reference Cheat Sheet

Concept	Formula / Code	Yaad Rakhna
Margin Width	2/	
SVM Objective	min 0.5	
RBF Kernel	exp(-gamma *	
C (soft margin)	Bigger C = tighter fit	C=1 se start karo
Gamma	Bigger = complex boundary	'scale' default
Scaling	StandardScaler	ALWAYS karo
Kernel choice	Linear for text, RBF for rest	Practical rule

Purana Tarika vs Naya Tarika

Task	Purana	Naya (sklearn 1.0+)
Confusion Matrix	sns.heatmap(cm, annot=True)	ConfusionMatrixDisplay.from_estimator()
ROC Curve	manual roc_curve + plt.plot	RocCurveDisplay.from_estimator()
Hyperparameter tuning	Manual loops	GridSearchCV

Common Mistakes to Avoid

- Scaling bhool jana (SVM distance-based hai!)
- Probability=True set nahi karna (ROC ke liye chahiye)
- C aur gamma dono ek saath bahut bada karna (overfit)
- GridSearchCV mein scoring=recall nahi dena (medical data ke liye)
- Test data par scaler.fit_transform karna (data leakage!)

8. SVM Capstone Project

Project: Medical Diagnosis System

Dataset: Breast Cancer Wisconsin (sklearn built-in) **Goal:** Malignant vs Benign classify karo with maximum Recall

Part A: Kernel Battle

1. Linear, RBF (gamma=scale), RBF (gamma=0.1), Polynomial (d=2,3), Sigmoid compare karo
2. Accuracy, Precision, Recall, F1 ek table mein dikhao
3. Medical context mein kaunsa best hai? Justify karo.

Part B: Hyperparameter Grid

1. GridSearchCV: C=[0.001, 0.01, 0.1, 1, 10, 100, 1000], gamma=['scale', 0.001, 0.01, 0.1, 1]
2. scoring='recall' use karo
3. Heatmap: C vs Gamma vs Recall score

Part C: Feature Engineering

1. Top 10 features (by correlation) vs full 30 features
2. PCA se 10 components extract karo aur compare karo
3. Kaunsa approach best tha?

Part D: Deploy-Ready Pipeline

1. Best model save karo: `import joblib; joblib.dump(svm, 'cancer_svm.pkl')`
2. Load karke naye patient par predict karo
3. Confidence score bhi dikhao

Bonus:

- class_weight='balanced' try karo — kya recall improve hota hai?
- Precision-Recall curve plot karo (ROC se compare karo)
- SVM vs Logistic Regression vs KNN compare karo ek final table mein

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Atlas Skilltech University — Machine Learning Course SVM Theory Guide — Hinglish Edition